

# NTAP, Coding, and DRG Analysis with Cost Scenarios

Prepared for direct client use. Facts are source-backed; inferences are clearly labeled.

## 1) NTAP Submission Form / Template

CMS does not accept NTAP applications by email or offline form. Applications are submitted through MEARIS only. For offline preparation, CMS publishes FY-specific reference materials (application overviews, cost workbook templates, and tracking forms) in the FY 2027 NTAP resources package. [1][2][3]

Practical answer: use the FY 2027 resource package as the closest sample template, but treat MEARIS as the operative submission system. [2][3]

## 2) ICD-10-PCS Section X (New Technology) Coding Mechanics

Fact: In this context, 'X code' refers to ICD-10-PCS Section X new technology coding, not HCPCS/CPT Part B X modifiers. [24]

Fact: CMS coding guidance treats Section X as the framework for describing qualifying new technology procedures in the inpatient coding environment. [24]

Inference: The relevant question for Product X is whether it can be distinctly identified on the inpatient claim through an ICD-10-PCS strategy, not whether a specific existing Section X code already applies today. [1][7][24]

## 3) NTAP Pathway and Interim Market Behavior (Stroke/EVT Context)

| Milestone                       | Typical FY 2027 Timing   | Source |
|---------------------------------|--------------------------|--------|
| Application submission (MEARIS) | October 2025 cycle dates | [1]    |
| Town Hall (SCI discussion)      | December 2025            | [1]    |
| Supplemental info deadline      | December 15, 2025        | [1]    |
| Proposed rule discussion        | April 2026 window        | [1]    |
| FDA authorization deadline      | May 1, 2026              | [1]    |
| Final rule                      | August 1, 2026           | [1]    |
| NTAP effective date             | October 1, 2026          | [1]    |

Inference: Between launch and NTAP activation, hospitals usually face a budget-exposure period because incremental technology cost is carried inside fixed DRG reimbursement. This often leads to selective use in early adoption windows until payment alignment improves.

Inference: A DRG update is a plausible pathway contingent on distinct inpatient claim identification, sufficient claim experience, and a durable cost signal.

## 4) EVT Trends and DRG Impact (2020 onward)

Fact: The relevant DRG landing zone for Product X remains case-definition dependent and must be validated against actual ICD-10-CM/PCS coding logic and grouper results. The DRGs discussed here are directional comparators for payment adequacy analysis, not a final grouping determination.

Fact: CMS Table 5 files show annual DRG relative-weight updates for stroke-related DRGs. The table below anchors FY2020 to FY2026 changes. [13]-[19]

| DRG | FY2020 | FY2021 | FY2022 | FY2023 | FY2024 | FY2025 | FY2026 |
|-----|--------|--------|--------|--------|--------|--------|--------|
| 023 | 5.6171 | 5.6623 | 5.6719 | 5.7314 | 5.6688 | 5.7047 | 5.7303 |
| 024 | 4.0165 | 3.9325 | 3.9390 | 3.9488 | 3.7888 | 3.8014 | 3.9119 |

|     |        |        |        |        |        |        |        |
|-----|--------|--------|--------|--------|--------|--------|--------|
| 061 | 2.7935 | 2.8882 | 2.8912 | 2.9326 | 2.8028 | 2.7032 | 2.7571 |
| 062 | 2.0112 | 1.9872 | 1.9883 | 1.9172 | 1.8717 | 1.7808 | 1.7572 |
| 063 | 1.6808 | 1.7099 | 1.7097 | 1.5810 | 1.4868 | 1.4047 | 1.4038 |

DRG definitions and comparator roles used in this analysis

| DRG | Definition (FY 2026 title)                                                                                                 | Comparator role in this model |
|-----|----------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| 023 | CRANIOTOMY WITH MAJOR DEVICE IMPLANT OR ACUTE COMPLEX CNS PRINCIPAL DIAGNOSIS WITH MAJOR OR ANTI-EPILEPTIC DRUG IMPLANT OR | SWH Outpatient comparator     |
| 024 | CRANIOTOMY WITH MAJOR DEVICE IMPLANT OR ACUTE COMPLEX CNS PRINCIPAL DIAGNOSIS WITH MAJOR OR ANTI-EPILEPTIC DRUG IMPLANT OR | SWH Outpatient comparator     |
| 061 | ISCHEMIC STROKE, PRECEREBRAL OCCLUSION OR TRANSIENT ISCHEMIA WITH THROMBOLYTIC TREATMENT                                   | TOC Outpatient comparator     |
| 062 | ISCHEMIC STROKE, PRECEREBRAL OCCLUSION OR TRANSIENT ISCHEMIA WITH THROMBOLYTIC TREATMENT                                   | TOC Outpatient comparator     |
| 063 | ISCHEMIC STROKE, PRECEREBRAL OCCLUSION OR TRANSIENT ISCHEMIA WITH THROMBOLYTIC TREATMENT                                   | TOC Outpatient comparator     |

Year-over-year change

| DRG | 2020->2021 | 2021->2022 | 2022->2023 | 2023->2024 | 2024->2025 | 2025->2026 |
|-----|------------|------------|------------|------------|------------|------------|
| 023 | +0.8%      | +0.2%      | +1.0%      | -1.1%      | +0.6%      | +0.4%      |
| 024 | -2.1%      | +0.2%      | +0.2%      | -4.1%      | +0.3%      | +2.9%      |
| 061 | +3.4%      | +0.1%      | +1.4%      | -4.4%      | -3.6%      | +2.0%      |
| 062 | -1.2%      | +0.1%      | -3.6%      | -2.4%      | -4.9%      | -1.3%      |
| 063 | +1.7%      | -0.0%      | -7.5%      | -6.0%      | -5.5%      | -0.1%      |

Inference: The pattern is not a single directional step-change. These are annual relative weights, not payment rates, and they should be interpreted as directional context rather than definitive Product X grouping evidence.

Fact: A public JNIS analysis through 2021 reported a flattening growth curve in U.S. mechanical thrombectomy procedure counts versus earlier rapid expansion. [20]

Other Episode Costs and Payment Classification

Fact: The model separates bundled inpatient facility costs from the full-episode spending components payers track across the care continuum. The first three lines are the bundled DRG-margin bucket (\$20,000 subtotal); the remaining lines are non-facility episode costs outside the index DRG. [6][7][9][21]

| Bucket                                                           | Where it comes from                                                                     | How it is paid                                                                                          | Baseline amount |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------|
| ICU / neuro-ICU monitoring, nursing, telemetry                   | 2-4 ICU days and high-intensity nursing/monitoring in the attached Product X materials. | Bundled in MS-DRG facility payment under IPPS (with possible outlier support for very high-cost cases). | \$12,000        |
| Routine floor days, room and board, standard nursing             | Remaining non-ICU days in a 6-8 day typical stay.                                       | Bundled into the inpatient MS-DRG under IPPS; not a separate hospital line item.                        | \$5,000         |
| Ancillary hospital services, pharmacy, supplies, case management | Imaging, labs, pharmacy workflow, consumables, and discharge planning.                  | Bundled into the inpatient MS-DRG under IPPS.                                                           | \$3,000         |
| Physician professional services                                  | Neurology/interventional/professional claims during stay.                               | Paid separately under the Physician Fee Schedule / Part B.                                              | \$3,500         |
| Post-acute care after discharge                                  | SNF/IRF/home health/LTCH after discharge.                                               | Paid under separate PPSs outside the index DRG.                                                         | \$14,000        |
| Readmission and ED revisit (90-day window)                       | Unplanned post-discharge acute utilization and revisit burden.                          | Paid as separate claims outside the index DRG.                                                          | \$6,000         |
| Long-term support/disability-related care (expected value)       | Blended expected-value allocation for long-term support in poor recovery states.        | Paid across post-acute and community support channels (payer-dependent).                                | \$12,000        |

Inference: Inpatient DRG-margin modeling should use only the bundled facility subtotal. Commercial and total-cost-of-care decisions should use the full episode view, including professional, post-acute, readmission, and long-term support spend.

Fact: CMS's 3-day payment-window policy also requires certain outpatient diagnostic and admission-related services furnished immediately prior to admission to be included on the inpatient Part A bill, reinforcing that these costs flow into the inpatient facility payment context. [22]

Fact: Sponsor materials on file tie the operating burden to a 48-hour infusion, ICU-level monitoring, repeated bag changes, and pharmacy coordination, and they provide the inpatient-day and ICU-day planning anchors used here. [23]

- 023 is essentially flat across the window, with the only dip occurring in FY2024 and partial recovery thereafter.
- 024 declines in FY2024 and then posts the strongest FY2026 rebound among the five DRGs.
- 061 peaks in FY2023 and then softens through FY2025 before a modest FY2026 recovery.
- 062 and 063 show the clearest multi-year erosion, which is the strongest signal that the stroke mix is not static.
- The effect is drift, not collapse: DRG weights move within a narrow band, which matters for reimbursement timing even when the clinical case mix is changing.

## 5) Economic Frameworks

### Hospital Adoption Economics (MS-DRG / NTAP)

| Product X price | DRG payment | Bundled facility cost | No NTAP margin | NTAP amount | DRG margin with NTAP | Post-DRG margin (\$) |
|-----------------|-------------|-----------------------|----------------|-------------|----------------------|----------------------|
| \$25,000        | \$30,000    | \$20,000              | -\$15,000      | \$9,750     | -\$5,250             | \$5,000              |
| \$35,000        | \$30,000    | \$20,000              | -\$25,000      | \$16,250    | -\$8,750             | -\$5,000             |
| \$50,000        | \$30,000    | \$20,000              | -\$40,000      | \$26,000    | -\$14,000            | -\$20,000            |

Modeled NTAP = lesser of: (a) 65% x Product X cost, or (b) 65% x (total case cost - MS-DRG payment).

Worked example (Product X = \$25,000): total case cost = \$45,000, DRG payment = \$30,000, excess cost = \$15,000. 65% x product cost = \$16,250; 65% x excess cost = \$9,750; therefore NTAP = \$9,750.

Inference: Clinical value does not by itself remove adoption friction. The hospital adoption constraint remains DRG-margin pressure during the early period.

### Payer Full-Episode Economics (Total Cost of Care)

| Component                                   | Baseline amount |
|---------------------------------------------|-----------------|
| Index inpatient payment                     | \$30,000        |
| Physician professional services             | \$3,500         |
| Post-acute care                             | \$14,000        |
| Readmission and ED revisit                  | \$6,000         |
| Long-term support / disability-related care | \$12,000        |

| Scenario     | Physician red. | Post-acute red. | Readmission red. | Long-term support red. | Interpretation                               |
|--------------|----------------|-----------------|------------------|------------------------|----------------------------------------------|
| Conservative | 10.0%          | 20.0%           | 15.0%            | 25.0%                  | Modest LOS and downstream effect             |
| Base         | 20.0%          | 40.0%           | 30.0%            | 50.0%                  | Moderate LOS/discharge and post-acute effect |
| Optimistic   | 30.0%          | 70.0%           | 60.0%            | 100.0%                 | Modeled full long-term support avoidance     |

| Scenario     | Episode delta @ \$25K | Episode delta @ \$35K | Episode delta @ \$50K |
|--------------|-----------------------|-----------------------|-----------------------|
| Conservative | \$2,700               | \$9,200               | \$18,950              |
| Base         | -\$4,350              | \$2,150               | \$11,900              |
| Optimistic   | -\$16,700             | -\$10,200             | -\$450                |

Inference: Hospital adoption economics and payer full-episode economics should be assessed separately. A favorable total-cost-of-care signal does not by itself remove hospital DRG-margin constraints.

## References

- [1] CMS New Medical Services and New Technologies (NTAP) page: <https://www.cms.gov/medicare/payment/prospective-payment-systems/acute-inpatient-pps/new-medical-services-and-new-technologies>
- [2] FY 2027 NTAP application resources zip: <https://www.cms.gov/files/zip/fy2027ntapapplication-resources-reference-onlyzip.zip>
- [3] MEARIS public resources: <https://mearis.cms.gov/public/resources>
- [4] MEARIS public publications: <https://mearis.cms.gov/public/publications>
- [5] Process for requesting new/revised ICD-10-PCS procedure codes: <https://www.cms.gov/medicare/coding-billing/icd-10-codes/process-for-requesting-new-revised-icd-10-pcs-procedure-codes>
- [6] Acute Inpatient PPS landing page: <https://www.cms.gov/medicare/payment/prospective-payment-systems/acute-inpatient-pps>
- [7] CMS guide for medical technology companies and other interested parties - IPPS: <https://www.cms.gov/cms-guide-medical-technology-companies-and-other-interested-parties/payment/ipp>
- [8] CMS outlier payments page: <https://www.cms.gov/Medicare/Medicare-Fee-for-Service-Payment/AcuteInpatientPPS/outlier.html>
- [9] Physician Fee Schedule: <https://www.cms.gov/cms-guide-medical-technology-companies-and-other-interested-parties/payment/physician-fee-schedule>
- [10] Prospective Payment Systems - General Information: <https://www.cms.gov/medicare/payment/prospective-payment-systems>
- [11] FY 2026 ICD-10-PCS code tables and index zip: <https://www.cms.gov/files/zip/2026-icd-10-pcs-code-tables-and-index.zip>
- [12] FY 2026 ICD-10-PCS codes file zip: <https://www.cms.gov/files/zip/2026-icd-10-pcs-codes-file.zip>
- [13] FY 2020 IPPS Table 5 zip: <https://www.cms.gov/medicare/medicare-fee-for-service-payment/acuteinpatientpps/downloads/fy2020-fr-table-5.zip>
- [14] FY 2021 IPPS Table 5 zip: <https://www.cms.gov/files/zip/fy-2021-ipp>
- [15] FY 2022 IPPS Table 5 zip: <https://www.cms.gov/files/zip/fy2022-ipp>
- [16] FY 2023 IPPS Table 5 zip: <https://www.cms.gov/files/zip/fy2023-ipp>
- [17] FY 2024 IPPS Table 5 zip: <https://www.cms.gov/files/zip/fy2024-ipp>
- [18] FY 2025 IPPS Table 5 zip: <https://www.cms.gov/files/zip/fy-2025-ipp>

[19] FY 2026 IPPS Table 5 zip: <https://www.cms.gov/files/zip/fy2026-ipp-5-table-5.zip>

[20] JNIS 2023 thrombectomy trend analysis: <https://jnis.bmj.com/content/15/e3/e349>

[21] CMS SNF payment basics (post-acute separate PPS): <https://www.cms.gov/Outreach-and-Education/Medicare-Learning-Network-MLN/MLNProducts/EnrollmentResources/provider-resources/snf/>

[22] CMS 3-day payment window (outpatient services rolled into inpatient Part A bill):  
<https://www.cms.gov/medicare/payment/prospective-payment-systems/acute-inpatient-pps/three-day-payment-window>

[23] Open-access acute ischemic stroke cost reference: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11064842/>

[24] 2026 Official ICD-10-PCS Coding Guidelines:  
<https://www.cms.gov/files/document/2026-official-icd-10-pcs-coding-guidelines.pdf>